

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2


ST. ANDREW'S JUNIOR COLLEGE
2015 JC2 Preliminary Examinations

H2 BIOLOGY
9648/2

Paper 2: Core (Mark Scheme)

Wednesday

 2nd September 2015

2 hours

 Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all eight** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question only.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

INFORMATION TO CANDIDATES

At the end of the examination,

1. Attach Section B answers to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/10
2	/8
3	/7
4	/11
5	/11
6	/10
7	/12
8	/11
Total	/80

This document consists of **20** printed pages.

[Turn over]

Section A

Answer **all** questions.

1. Brown adipose tissue (BAT), one of two types of fat tissue found in mammals, is highly specialized for non-shivering thermogenesis. Fig 1.1 shows how mitochondria in BAT cells function differently from those in other cells during periods of cold environmental conditions.

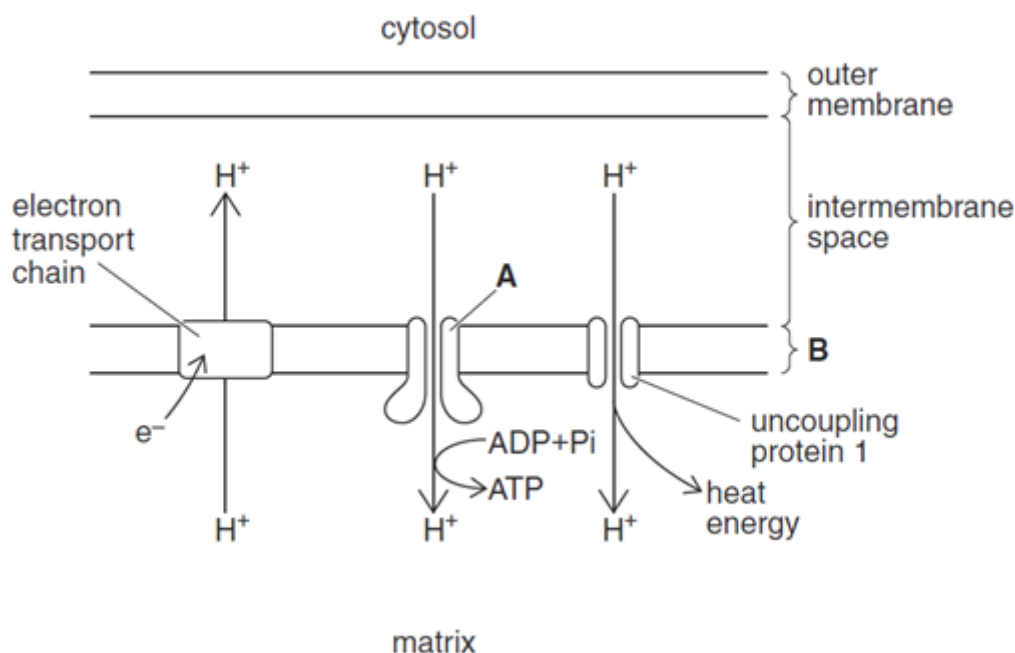


Fig. 1.1

(a) Name structures **A** and **B**.

A ATP synthase

B Inner membrane

[2]

(b) Describe where the electrons that are passed along the electron transport chain come from.

.....[2]

- 1 electron comes from hydrogen (atom)
- 2 from NADH / $FADH_2$
- 3 produced (during dehydrogenation reactions) in glycolysis, link reaction, Krebs cycle

(c) Suggest how the uncoupling protein 1 found in the mitochondria of BAT cells is involved in thermogenesis.

.....[1]

energy in the proton-motive force / proton gradient is converted to heat instead of being used for ATP

/ allows H^+ to diffuse down its concentration gradient, and produces heat energy.

/ it uncouples proton gradient and ATP synthesis

(d) Cyanide inhibits cytochrome c oxidase, which is a large transmembrane protein complex found in the electron transport chain (ETC) of mitochondria.

(i) Explain how this stops aerobic respiration.

.....[3]

[Any 2 points for 1 mark]

- 1 oxygen cannot accept electrons (and protons to form water);
- 2 electrons cannot be transported along ETC / ETC stops working
- 3 proton gradient cannot be maintained;
- 4 preventing oxidative phosphorylation / no ATP synthesis
- 5 NAD^+ / FAD cannot be regenerated;
- 6 preventing link reaction and Krebs cycle;

(ii) Explain how BAT cells can still generate ATP in the presence of cyanide.

.....[2]

- 1 BAT cells (undergo anaerobic respiration where) pyruvate is reduced/converted to lactate (by NADH);;
- 2 NAD^+ is regenerated for use in glycolysis to continuously generate ATP;;

[Q1 Total: 10]

2. (a) *Hibiscus tiliaceus* is a common coastal plant found on the wild shores of Singapore. Fig. 2.1 shows how the amount of DNA per cell of *Hibiscus tiliaceus* varies during periods A to E in the cell cycle.

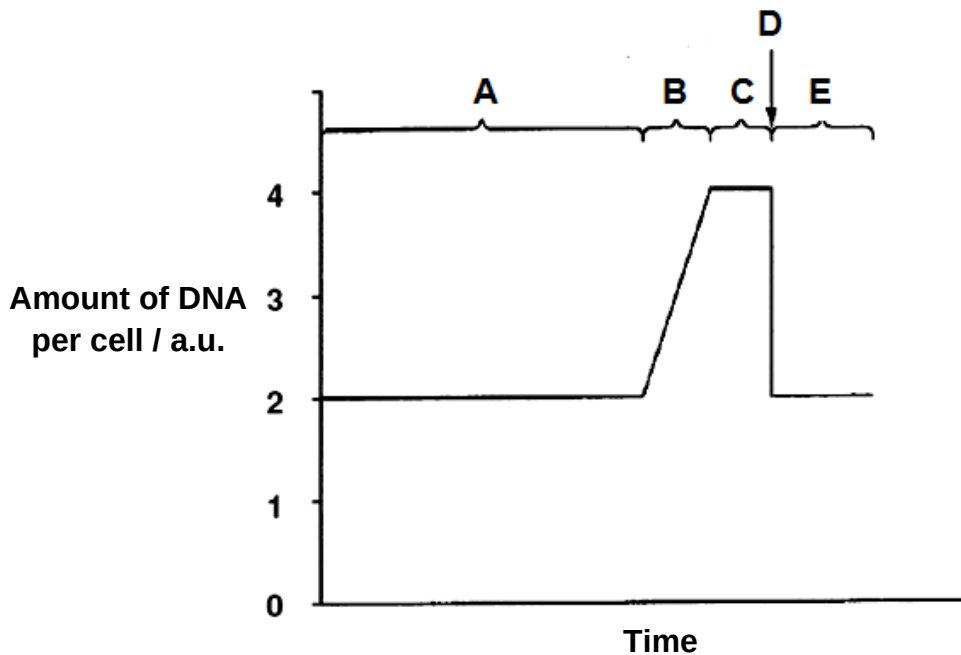


Fig. 2.1

Radioactive thymine was supplied to the cells of some growing *Hibiscus* tissue.

(i) Identify the period (A to E) during which anaphase occurs. Explain your answer.

.....[3]

- 1 Period C
- 2 (semi-conservative) DNA replication has occurred, DNA amount has doubled / increased from 2 a.u. to 4 a.u.
- 3 cytokinesis has not occurred (to halve DNA amount per cell)
/ separation of sister chromatids during anaphase occurs within the same cell

Fig. 2.2 shows representations of the stages of meiosis, **A** to **J** in *Hibiscus tiliaceus*, but they are not in the correct order.

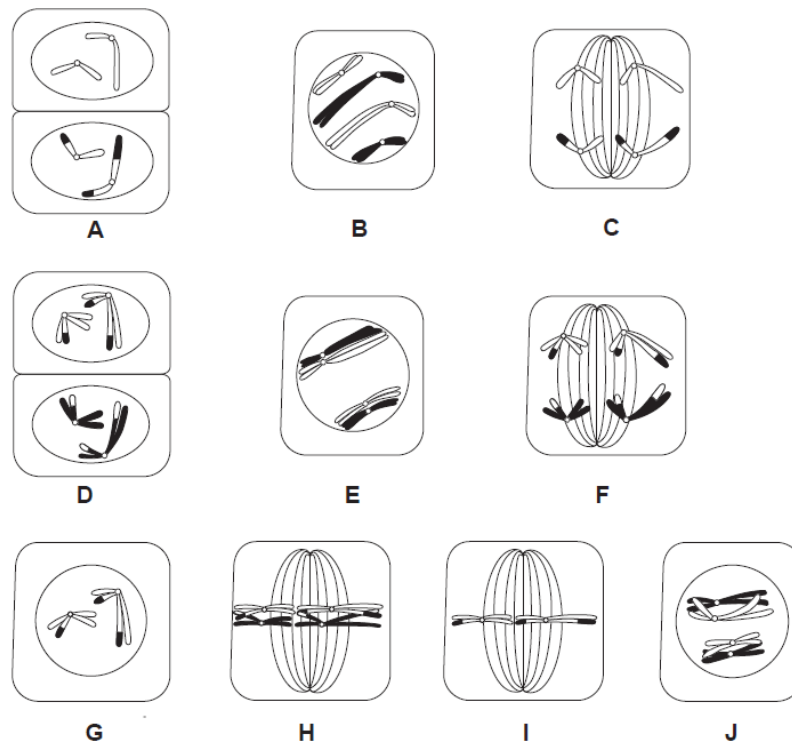


Fig 2.2

(ii) Complete the table below by writing the stages of meiosis in the correct order. Some of the stages have already been written in the table.

Meiosis I

B	E	J	H	F	D
----------	----------	----------	----------	----------	----------

[1]

Meiosis II

G	I	C	A
----------	----------	----------	----------

[1]

(iii) Explain how meiosis can result in genetic variation amongst offspring of *Hibiscus tiliaceus*.

.....[3]

- 1 crossing over between non-sister chromatids of homologous chromosomes in prophase I ;
- 2 independent assortment in metaphase I/II
/ random arrangement and subsequent separation of **homologous chromosomes** at metaphase I and anaphase I respectively ; / random arrangement and subsequent separation of **chromatids** at metaphase II and anaphase II respectively ;
- 3 results in new combination of alleles ;

3.(a) In an investigation of DNA replication, a species of bacterium was grown for many generations in a medium containing ^{15}N isotope of nitrogen.

The culture was then switched to a medium containing only ^{14}N for one generation of growth; it was then returned to a ^{15}N -containing medium for two more generations of growth. DNA was then isolated from the bacterial cells, and the density of the DNA was measured using density gradient centrifugation.

In this technique, samples of different densities settle at different levels in the tube. Fig. 3.1 shows the positions of ^{15}N - and ^{14}N -labeled DNA respectively.

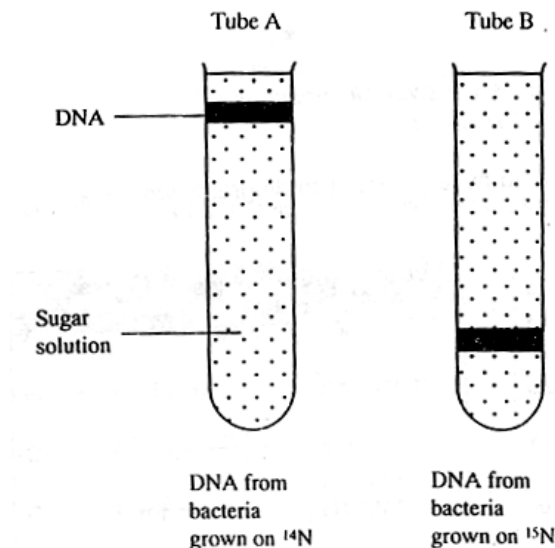


Fig. 3.1

(i) State which part of the nucleotide contains nitrogen.

.....[1]
(purine/pyrimidine/nitrogenous) bases

(ii) In Fig. 3.2, draw and label the pattern of the possible band(s) that can be obtained at the end of the two more generations of growth in the ^{15}N -containing medium.

[2]

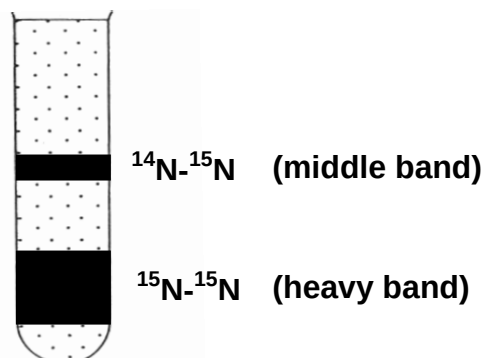


Fig 3.2

- 1 2 bands at correct positions with labels – 1 at mid position, 1 at bottom position
- 2 correct relative thickness – heavy band approx. 3x thicker than middle band

(b) The ribosomal RNA genes in human cells are located near the tips of five different chromosomes (13, 14, 15, 21 and 22). Ten small nucleoli form after mitosis in a human cell and then quickly grow and fuse to form a single large nucleolus.

Fig. 3.3 shows diagrammatically the relationship between the DNA of the ribosomal RNA genes and the nucleolus.

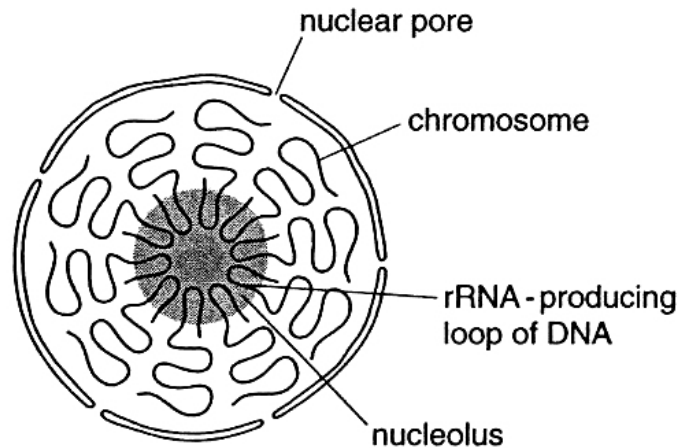


Fig. 3.31

(i) With reference to Fig. 3.3, explain why ten small nucleoli form.

.....[1]
 human cells are diploid
 / two homologous chromosomes of each type
 / five chromosomes would exist in pair
 / there are 10 chromosomes

(ii) State the function of the nuclear pores in this context.

.....[1]
 allow passage of ribosomal **subunits** out of the nucleus / into the cytoplasm
 / ribosomal **proteins** into the nucleus

Reject: to allow rRNA to leave the nucleus. (rRNA does not leave the nucleus).

(iii) Suggest why the nucleolus is not present during mitosis.

.....[2]
 1 nucleolus is site of rRNA synthesis;
 2 no transcription occurs during mitosis
 / there is no rRNA synthesis during mitosis
 3 rRNA genes are condensed into chromosomes

Examiners comments: Protein synthesis (transcription and translation processes) only occur during G1 and G2 of interphase.

[Q3 Total: 7]

4. Fig 4.1 shows a T4 bacteriophage virus infecting a bacterium.

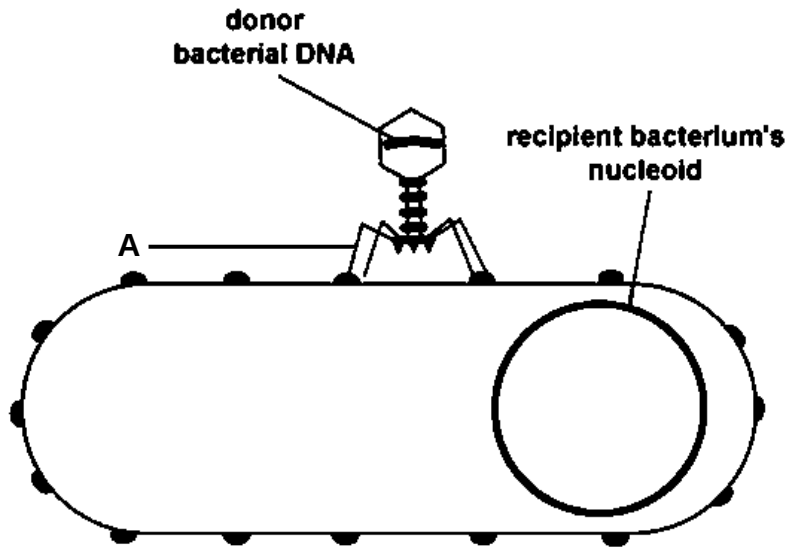


Fig. 4.1

(a)(i) Explain how **structure A** facilitates the bacteriophage's infection process in Fig. 4.1.

-[1]
- 1 presence of attachment sites on tail fibres / structure A;
recognise and bind / adsorb to complementary/specific receptors on bacterial cell (wall);

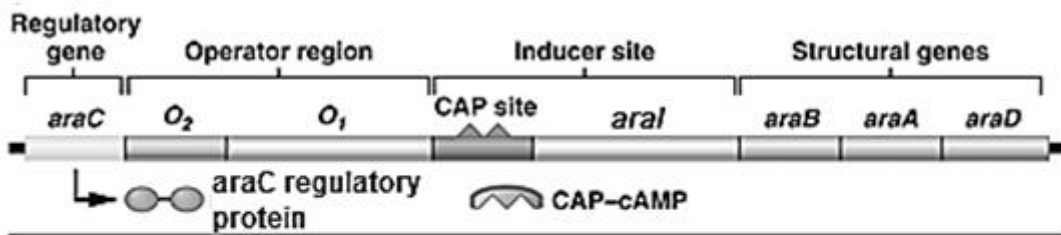
(ii) Describe how the bacteriophage in Fig. 4.1 facilitates the transfer of antibiotic resistance genes in a bacterial population.

-[3]
- 1 Phage undergoes generalised transduction (during the lytic cycle);;
 - 2 **Donor** bacteria DNA is hydrolysed into fragments;;
 - 3 (Random) packaging of host **antibiotic resistance genes** into the capsid of newly assembled phages;
 - 4 Phage released can **infect another** bacterium, transferring the antibiotic resistance genes via **genetic/homologous recombination / insertion / integration**;
(expression of antibiotic resistance genes result in production of antibiotic resistance proteins in the newly infected cell)

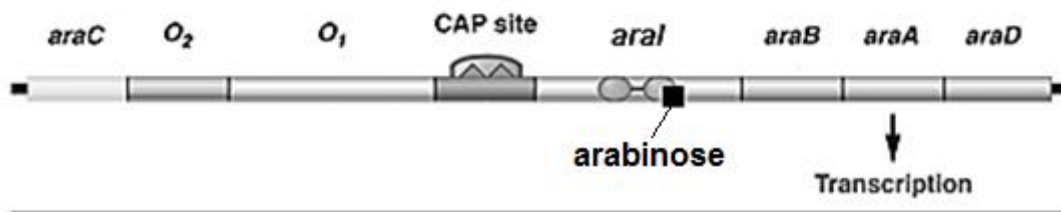
Fig. 4.2 shows the arabinose operon in *E.coli*. The arabinose operon has three structural genes, *araB*, *araA* and *araD*, which encode for enzymes needed for the catabolism of arabinose, a five carbon sugar that can be used by *E.coli* as an alternative carbon source. The operon has both positive and negative regulation, being activated in the presence of arabinose.

There are two operator sites (O_1 & O_2), the promoter site (situated in *araI*), and the *araC* gene which encodes for the AraC regulatory protein. The CAP protein functions the same way as in the *lac* operon.

(a) The arabinose operon with its regulatory gene



(b) Positive regulation of arabinose operon



(c) Negative regulation of arabinose operon

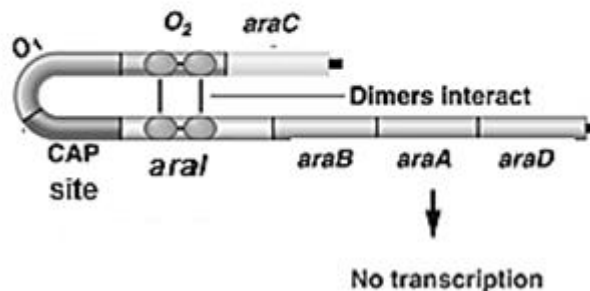


Fig. 4.2

With reference to Fig. 4.2,

(b)(i) describe and explain the **positive** regulation of the *ara* operon.

.....[3]

- 1 CAP-cAMP binds to the CAP site (Accept CAP-binding site);
- 2 Together with the binding of **arabinose-bound *araC* regulatory protein** at the ***araI* site/ inducer site**;
- 3 This facilitates efficient positioning of RNA Polymerase at the promoter / prevents *araC* protein/dimer interaction / looping of DNA which allow access of RNA polymerase to structural genes, (leading to high rate of transcription of structural genes);

ii) Contrast the **negative** regulation of the *ara* operon with that of the *lac* operon.

.....[2]

<i>ara</i> operon	<i>lac</i> operon
1 In the absence of arabinose, <u>AraC regulatory protein</u> (unbound by arabinose) recognized and bind to both <u><i>araI</i> & <i>O</i>₂ sites</u>	In the absence of lactose, lac <u>repressor</u> recognizes and binds to <u>operator</u> ,
2 resulting in a DNA loop / araC protein/dimer interaction that prevents RNA polymerase from binding to promoter / transcribing the structural genes;;	Block/overlaps with the promoter; prevents RNA polymerase from binding to promoter / transcribing the structural genes;;

Examiner's comments:

Reject answers merely relating to structural differences e.g. comparing the number of operators present without mentioning about the process of regulation.

(iii) A deletion occurred at the beginning of the ***araA*** gene. Predict the effect of this deletion on the gene product of the downstream structural gene, ***araD***. Explain.

.....[2]

- 1 [Predict] Gene product of *araD* remains functional / no change in tertiary structure
- 2 [Explain] Gene sequence of *araD* codes for its own start and stop codons (and is thus, not affected by the deletion of the upstream *araA* gene);;

[Q4 Total: 11]

5.(a) Paramyotonia congenita is a genetic disease that involves muscle stiffness and weakness. One of the causes of the disease is a single base substitution in the SCN4A gene.

The SCN4A gene codes for the alpha subunit of Na⁺ channels that are abundant in skeletal muscles. The mutant SCN4A protein is full-length and permits Na⁺ entry like a normal protein channel would, but the protein gate is unable to be inactivated as quickly as a normal Na⁺ channel.

Suggest possible reasons for the mutated protein to remain full-length and retain the same basic function as the normal protein.

.....[3]

- 1 no frame-shift mutation, number of amino acids coded for remains the same
- 2 substituted amino acid has similar properties as the original one, similar tertiary structure
- 3 mutation results in change of amino acid in a region (of the primary sequence) which are not essential for the protein function

Examiners' comments

REJECT: "Degeneracy of genetic code / substitution of single base may still code for same amino acid." In the question, a "mutant protein" is produced. That means that its amino acid sequence is different from the normal protein.

(b) Double-stranded DNA of the SCN4A gene is denatured and allowed to hybridise through complementary base pairing with mature SCN4A mRNA isolated from skeletal muscle cells. This is shown in Fig. 5.1.

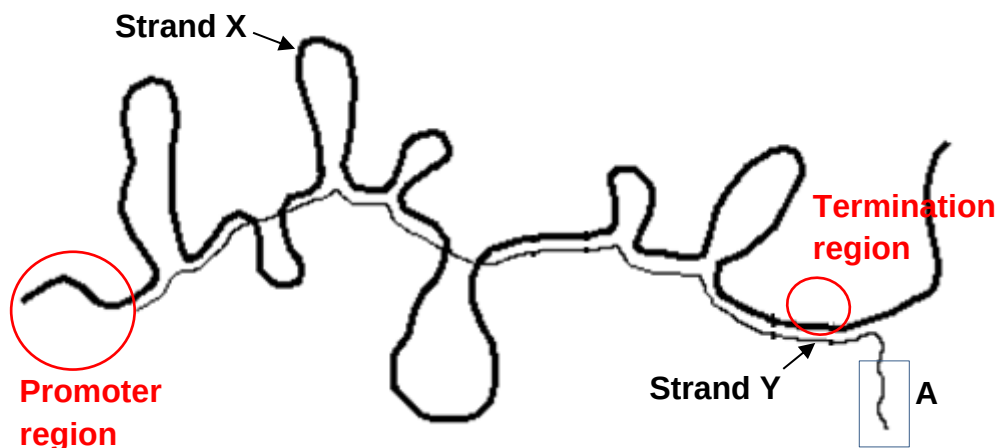


Fig 5.1

(i) State the identity of Strand X.

.....[1]

Template DNA strand / single-stranded DNA (of SCN4A gene)

REJECT: DNA (which is double-stranded)

(ii) Explain your answer in (i) using evidence from Fig. 5.1.

.....[3]

- 1 Strand X is longer than Strand Y
- 2 Strand X longer due to introns OR the looped portions consist of introns
OR Strand Y shorter due to introns being excised (to form the mature mRNA)
/ Strand X longer due to promoter / terminator OR Strand Y shorter as promoter /
terminator not transcribed
- 3 No **complementary** sequence on Strand Y (mature mRNA) for complementary
base pairing with non-coding sequences in Strand X
/ Strand X forms **complementary** base pairing at coding regions / UTRs on
Strand Y

(iii) Circle the locations of the promoter and termination regions of the SCN4A gene on Fig. 5.1. Label the regions accordingly. **Ans in RED.** [2]

(iv) What does the unpaired segment (labelled A) on strand Y represent? Explain your answer.

.....[2]

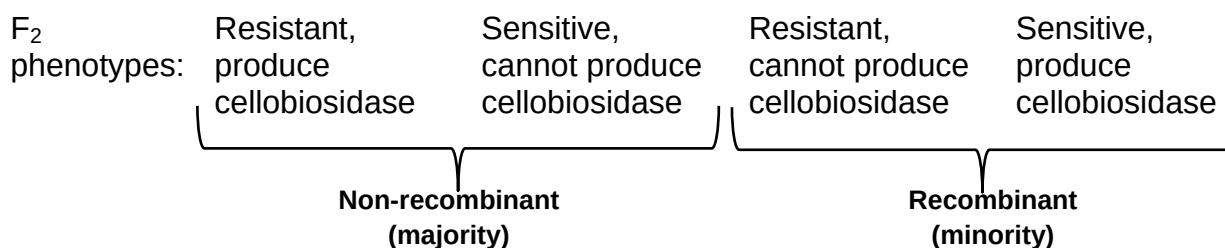
- 1 Poly (A) tail
- 2 Poly (A) tail is added to strand Y post-transcriptionally
/Template DNA does not code for the poly (A) tail

[Q5 Total: 11]

	<u>AB</u>	<u>ab</u>	<u>Ab</u>	<u>aB</u>
<u>AB</u>	$\frac{\underline{AB}}{AB}$ □	$\frac{\underline{AB}}{ab}$ □	$\frac{\underline{AB}}{Ab}$ □	$\frac{\underline{AB}}{aB}$ □
<u>ab</u>	$\frac{\underline{ab}}{AB}$ □	$\frac{\underline{ab}}{ab}$ ○	$\frac{\underline{ab}}{Ab}$ △	$\frac{\underline{ab}}{aB}$ ♥
<u>Ab</u>	$\frac{\underline{Ab}}{AB}$ □	$\frac{\underline{Ab}}{ab}$ △	$\frac{\underline{Ab}}{Ab}$ △	$\frac{\underline{Ab}}{aB}$ □
<u>aB</u>	$\frac{\underline{aB}}{AB}$ □	$\frac{\underline{aB}}{ab}$ ♥	$\frac{\underline{aB}}{Ab}$ □	$\frac{\underline{aB}}{aB}$ ♥

Key

- : Resistant to *Hexamermis*, produce cellobiosidase
 △ : Resistant to *Hexamermis*, cannot produce cellobiosidase
 ♥ : Sensitive to *Hexamermis*, produce cellobiosidase
 ○ : Sensitive to *Hexamermis*, cannot produce cellobiosidase



Mark scheme:

- 1 F₁ genotype (with linkage bar, e.c.f subsequently)
- 2 F₁ gametes
- 3 F₂ genotypes
- 4 F₂ genotypes correspond to phenotypes
- 5 Correct presentation of genetic diagram; indication of recombinant and non-recombinant gametes; indication of recombinant and non-recombinant phenotypes

Table 6.1 shows a chi-square table.

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(ii) Calculate the χ^2 value.

.....[1]
 659.40 (2 d.p.)

Examiners' comments: Follow the d.p. or the s.f. of the chi-squared table provided. In this case, the chi-squared table values are consistent with 2 d.p, NOT 3 s.f.. May be different in other cases!

(iii) Using your answer in b(ii) and Table 6.1, state what conclusion may be drawn from the result.

.....[3]

No marks awarded if calculated χ^2 value is incorrect.

- 1 Since the calculated χ^2 value 659.40 > critical χ^2 value 7.82 at $p = 0.05$;
/ At χ^2 value of 659.40, p value < 0.001, which is smaller than 0.05
- 2 The results of the χ^2 test suggest that there is a significant difference between the observed and the expected values (at the 5% level)
/ Any difference between the observed and the expected values is not due to chance
- 3 The predicted phenotype ratio of 9:3:3:1 is incorrect
/ There is no independent assortment of the two genes

[Q6 Total: 10]

7(a)(i) Explain the significance of the mitochondria on the maintenance of resting potential (-70 mV) in neurones.

-[1]
- 1 Synthesis of ATP (which powers the Na^+/K^+ pumps) to **pump 3 Na^+ out and 2 K^+ into the neurone** for maintenance of resting potential;;

Reject: hydrolysis of ATP by mitochondria

Fig. 7.1 shows the electron micrograph of a junction between two neurones where neurotransmitter dopamine are released.



Fig. 7.1

(ii) Identify the structures labelled A and B. [1]

- A: Synaptic vesicle
B: Synaptic cleft

Abnormally high dopaminergic synaptic transmission has been linked to psychosis. Anti-psychotic medications are available to block the effects of dopamine.

(iii) Suggest two possible mechanisms of such medications. [2]

- 1 Binds to dopamine receptors on post-synaptic membrane and prevent dopamine from binding. (e.c.f for missing "post-synaptic membrane")
- 2 Prevents the packaging of dopamine into synaptic vesicles
- 3 Prevents exocytosis of synaptic vesicles
- 4 Speeds up reuptake of dopamine into pre-synaptic neurones
- 5 Binds to dopamine and prevents it from binding to dopamine receptors on post-synaptic membrane.

[Any 2]

Heart rate is controlled by the two branches of the autonomic (involuntary) nervous system: the sympathetic nervous system (SNS) and the parasympathetic nervous system (PNS). The PNS releases the hormone acetylcholine to slow the heart rate.

M2 muscarinic receptors (a part of the PNS) belong to the GPCR family. They open up K^+ channels in the heart, allowing for K^+ efflux and contributing to slowing down of heart rate. K^+ channels consist of GIRK1 and GIRK4 subunits.

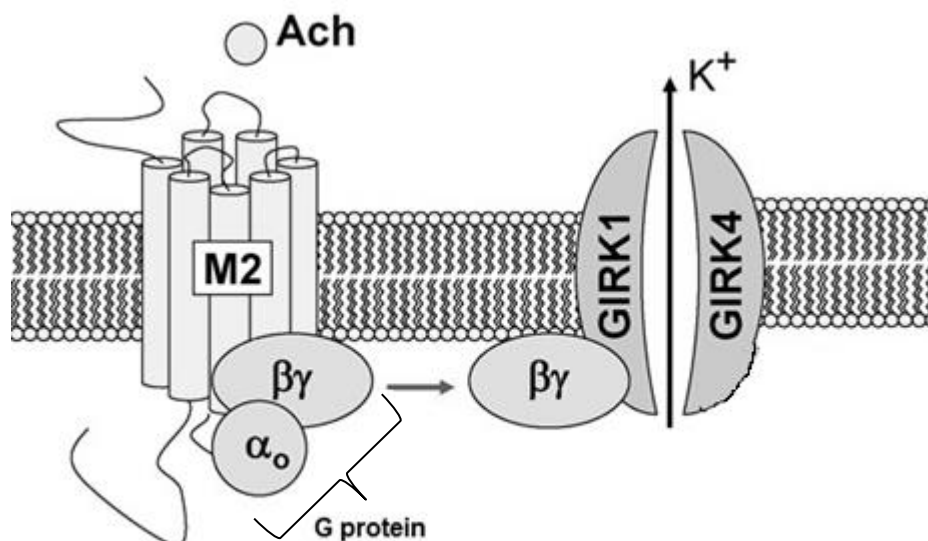


Fig. 7.2

(b)(i) Explain how the structure of the M2 muscarinic receptor allows it to be held in the membrane.

-[2]
- 1 Non-polar / hydrophobic R groups / side chains of amino acids of proteins
Interact with hydrophobic core / fatty acid tails of membrane;
 - 2 Polar / charged / hydrophilic R groups / side chains of amino acids of proteins
interact with hydrophilic phosphate head of membrane

Examiner's comments:

REJECT "hydrophilic/hydrophobic amino acids". It is the R groups of amino acids that are hydrophobic/hydrophilic.

(ii) Describe how acetylcholine slows down the heart rate.

.....[4]

- 1 Binding of acetylcholine triggers **conformational changes** which exposes the cytosolic G-protein binding domain of the GPCR, (activation of GPCR)
- 2 G protein binds to GPCR; the $\beta\gamma$ subunit of G protein dissociates from the receptor,
- 3 **moves/diffuses** along the cell surface membrane and binds to K^+ channels at the GIRK1 subunit

(K^+ ion channels open, diffusion of K^+ ions **out** of post-synaptic cell)

- 4 results in membrane hyperpolarization;
Harder for membrane potential to reach threshold potential;
/ **less** action potential generated in post-synaptic membrane (slower contractions);

(c) Blood glucose level is tightly regulated in the body. Other than the changes in enzymatic activities of glycogen synthetase and glycogen phosphorylase, state two other changes that will occur within cells possessing insulin receptors when the insulin binds to the receptors.

.....[2]

- 1 Increase uptake of glucose into cells / increase number of glucose transporters;
- 2 Increase in rate of conversion of glucose to fat (in adipose tissues) ;
- 3 Increase in rate of respiration / glucose oxidation;

[Any 2]

AVP:

1. Increases the synthesis of DNA and RNA.
/ Increases the synthesis of proteins from amino acids.

[Q7 Total: 12]

8. *Astyanax mexicanus* is a single species of tetra fish consisting of a sighted surface-dwelling form (surfacefish) and many blind cave-dwelling forms (cavefish) from different caves. The cavefish have small eyes that do not work, but are able to navigate in the complete darkness of the caves by emitting pressure waves with their mouths. It was found that the surfacefish were more heavily parasitized by a bacteria which infected the eyes as they do not have a thick layer of scale-like tissue covering their eyes, which is present in the cavefish.

It was thought that the ancestral morph of *Astyanax mexicanus*, which lived some 1.2 million years ago, had functional eyes. They were slowly displaced from their river habitats by large predatory fish into smaller streams that flowed through caves.

(a) Explain how natural selection could have resulted in the evolutionary change demonstrated here.

-[5]
- 1 Genetic variation in eye development is present in the ancestral morph due to mutation.
 - 2 When they were displaced into the smaller streams in caves, sight is no longer important.
/ Eyes became vestigial structures in the cavefish.
 - 3 The **selection pressure** is parasitism by bacteria
 - 4 Fish with thicker scale-like tissue covering their eyes were at a selective advantage, as they were less parasitized (accept converse argument)

(Award point 5 if correct idea is present in point 4).

- 5 Thus, **better able to survive to reproductive age and produce more offspring, passing down their alleles** (which determines the development of their eyes) to their offspring.
- 6 After many generations, natural selection will lead to change in frequency of alleles responsible for eye development.

OR

- 1 Genetic variation in the ability to emit pressure waves with mouths exists due to mutation.
- 2 Complete darkness in the caves exert a **selection pressure**
- 3 Fish able to navigate in the complete darkness of the caves by emitting pressure waves with their mouths were at a selective advantage as they are able to find food/ escape from predators better (accept converse argument)

(Award point 5 if correct idea is present in point 4).

- 4 Thus, **better able to survive to reproductive age and produce more offspring, passing down their alleles** (which determines the development of their eyes) to their offspring.
- 5 After many generations, natural selection will lead to change in frequency of alleles responsible for eye development.

(b) The evolution of the *Astyanax mexicanus* was demonstrated by anatomical homology. Using a named example, discuss another type of homology evidence supporting natural selection.

.....[4]

- 1 Embryological homology
/ Comparison of early developmental stages of organisms revealed that embryos from different species retain similarities during their development.
- 2 Such similarities/homology suggests inheritance from a common ancestor
- 3 which can later develop into homologous structures with very different functions, implying on descent with modification.
- 4 E.g. presence of a tail and pharyngeal pouches in all vertebrate embryos, which later develop gill slits in fishes and parts of ears and throat in humans.

OR

- 1 Molecular homology
/ Comparison between organisms at the molecular level revealed similarities
- 2 Such similarities/homology suggests inheritance from a common ancestor
- 3 As descendents of the same species evolved independently, more and more differences are accumulated in their DNA.
- 4 E.g. similarities in the sequence of DNA, RNA and amino acids of different organisms of common genes.

(c) A fossilised *Astyanax mexicanus* specimen that dated back to 0.8 million years ago was found in the subsidiary streams of the Rio Grande. Its DNA was extracted, sequenced and compared to those obtained from the surfacefish and cavefish. It was this evidence that led scientists to classify the three members as a single species.

Explain two advantages of using molecular methods in classifying organisms.

.....[2]

- 1 Quantifiable and open to statistical analysis;
Large quantities of data are required for statistical analysis; however there is little morphological data available
- 2 Unambiguous and objective;
Morphological data may differ depending on the way in which it was classified
- 3 Not affected by convergent evolution / some characteristics may be analogous; similar morphology may not have been inherited from common ancestor

Additional points

- 4 Neutral mutations are accumulated at a relatively constant rate for any particular gene and use as a molecular clock to date evolutionary events

UCLES Examiners' Comments

Answers to this question from some candidates were sometimes lacking in **precise terminology**. Points sought included the idea that molecular evidence was unambiguous and objective, that it was quantifiable and was open to statistical analysis. While morphological evidence could be due to convergence. More able candidates went on to explain the molecular clock idea and silent genes.

[Q8 Total: 11]

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 9** **(a)** Describe the molecular structure of glycogen and relate its structure to its function in living organisms. [6]
- (b)** Outline of the roles of cholesterol, glycolipids and glycoproteins in the cell surface membrane. [6]
- (c)** Describe how one bacterial cell can give rise to genetically identical cells in a colony. [8]

[Total: 20]

OR

- 10** **(a)** Using a named example, discuss how genetic variation may be preserved in natural population by heterozygote advantage. [8]
- (b)** Explain how biogeography supports the evolutionary deductions based on homologies. [6]
- (c)** Explain the conditions which are necessary for speciation to occur. [6]

9(a) Describe the molecular structure of glycogen and relate its structure to its function in living organisms. [6]

Structure (max 2)

- 1 glycogen is extensively branched
- 2 polymer of α -glucose molecules linked by $\alpha(1, 4)$ glycosidic bonds and $\alpha(1,6)$ glycosidic bonds;
- 3 coiled into a helix

Functions (max 4)

- 4 insoluble in water due to hydrophilic hydroxyl groups of glucose residues project into interior of helices;
- 5 stored in large quantities without having any great effect on the water potential of cells / prevented from diffusing out of cells.
- 6 no cross-linking between chains
/ no cross-linking to form big bundles
/ individual chains can fold into compact shape;
- 7 can be stored in large quantities within the limited space in cells
- 8 easily broken down to monosaccharides when required;

9(b) Outline the roles of cholesterol, glycolipids and glycoproteins in the cell surface membrane. [6]

- 1 Cholesterol is important for regulating/maintaining the fluidity of cell membranes.
- 2 at moderate/high temperatures, cholesterol reduces membrane fluidity by reducing phospholipids movement,
- 3 at low temperatures, it hinders solidification by disrupting the regular packing of phospholipids.
- 4 Glycolipids and glycoproteins are phospholipids and proteins respectively with a carbohydrate chain associated on the outer surface of the membrane
- 5 help maintain stability of the membrane by forming hydrogen bonds with water molecules surrounding the cell.
- 6 involved in cell recognition
- 7 and cell attachment/adhesion
- 8 recognition sites for antibodies
- 9 Glycoproteins act as cell surface receptors

max 6

9(c) Describe how one bacterial cell can give rise to genetically identical cells in a colony. [8]

Binary fission

- 1 Single double-stranded DNA chromosome attaches to the plasma membrane, at the origin of replication;
- 2 DNA replicates via bidirectional replication;
- 3 Ref. semi-conservative replication ;
- 4 At each replication fork, one new strand is made continuously as a leading strand, and the lagging strand is made discontinuously via the formation of Okazaki fragments;
- 5 Enzymes cut, twirl and reseal the double helix to prevent the DNA from tangling;
- 6 The newly synthesized DNA's origin of replication also attaches to the plasma membrane;
- 7 Bacterial cell then elongates, and the two bacterial chromosomes' origin of replication ends up in opposite poles of the cell;
- 8 The plasma membrane invaginates, pinching off the two halves of the original cell;
- 9 New cell wall material is then synthesized (giving rise to 2 genetically identical cells);

10(a) Using a named example, discuss how genetic variation may be preserved in natural population by heterozygote advantage. [8]

Hb^SHb^S individuals

- 1 Individual homozygous for Hb^S / genotype Hb^SHb^S suffer from sickle cell anaemia
- 2 the normal haemoglobin (haemoglobin A) in red blood cells is replaced entirely by abnormal haemoglobin (haemoglobin S).
- 3 Sickled red blood cells tend to stick together and obstruct blood flow in capillaries, depriving multiple organs of oxygen, resulting in organ damage / Sickled red blood cells rupture easily, resulting in anaemia and fatigue.
- 4 May result in early death. Therefore, there is a higher potential for the removal of the Hb^S allele from the gene pool.

Hb^AHb^A individuals

- 5 Hb^AHb^A individuals are at greater risk of dying of malaria compared to heterozygotes. Therefore, there is a higher potential for the removal of the Hb^A allele from gene pool.

Hb^AHb^S individuals

- 6 Individual who are heterozygous / genotype Hb^AHb^S suffer from sickle cell trait
- 7 produce both normal and abnormal haemoglobin
- 8 Hb^AHb^S individuals are a selective advantage and are able to survive and reproduce in malaria-infected areas.
- 9 (This condition is known as heterozygote advantage as) it preserves the Hb^S allele in the population
- 10 A mosquito transmits *Plasmodium*, the malaria parasite to humans. The malaria parasite spends part of its life cycle in red blood cells.
- 11 When malaria parasite invade the bloodstream, the red blood cell that contains haemoglobin S is sickled-shaped and is quickly destroyed by the body, trapping the parasites within the sickle cell and stopping the infection.
- 12 The slowdown in blood flow in sufferers of sickle cell trait also hampered the parasite's ability to travel and rapidly infect new cells.

10(b) Explain how biogeography supports the evolutionary deductions based on homologies. [6]

Biogeography

- 1 Biogeography refers to the study of the geographic distribution of species.
- 2 Biogeography suggests that species that are closely related tend to be found closer together and not found in similar environments far away due to geographical barriers.
- 3 E.g. Marsupials are not found anywhere else in the world except in Australia.
- 4 Biogeography also suggests that the organisms found close together are evolved from a common ancestor.
- 5 E.g. Island biogeography of the Galapagos: Most species of plants and animals living on the islands are closely related to neighbouring mainland (ancestral) species.
- 6 Homology observed between different island species suggests of evolution from mainland ancestors with the element of descent with modification.
- 7 E.g. different species of finches on Galapagos Islands exhibit homology in terms of shape and size of beak due to natural selection, suggesting of evolution from mainland ancestors.

10(c) Explain the conditions which are necessary for speciation to occur. [6]

- 1 Genetic variations should exist in the population; due to mutation;
- 2 Presence of reproductive isolation; that disrupts gene flow;
- 3 Geographical isolation due to physical barriers that separates two sub-populations;;
- 4 Physiological Isolation which occurs when female and male sex organs are not compatible;;
- 5 Behavioural Isolation which results in little or no sexual attraction between females and males;;
- 6 Temporal Isolation which occurs when sub-populations breed at different seasons or times of the day;;
- 7 Different selection pressures are exerted on the sub-populations, favouring different groups of organisms which are at selective advantage;
- 8 Mutations must occur to create (more) new alleles;;